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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

VENDOR SUPPLY TRACKING AND ORDER PROCESSING SYSTEM

Submitted by

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DECLARATION

I hereby declare that the project entitled " **Salary Payroll Processing and Payslip Generation System"** submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **MUGILAN R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(Jackvin J A)

CERTIFICATE

This is to certify that the project entitled **" Salary Payroll Processing and Payslip Generation System"** is a **bona fide work** carried out by **JACKVIN PAUL J A B.Sc.IT 'B' (2025-2028)**, **INFORMATION TECHNOLOGY**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:COIMBATORE

Date:

Submitted for the Viva-Voce Examination held on _____

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ABSTRACT

The **Salary Payroll Processing and Payslip Generation System** is a Java-based application developed to automate and simplify the process of managing employee salaries in an organization. Payroll management is a critical function in every business, as it involves calculating employee earnings, handling deductions, and ensuring timely salary distribution. Traditional payroll systems, which rely on manual calculations or basic spreadsheets, are often time-consuming, prone to errors, and difficult to manage as the number of employees increases.

This system provides an efficient and user-friendly solution by automating salary calculations and generating accurate payslips. It allows users to input employee details such as name, ID, and salary components including basic pay, allowances (such as HRA and DA), and deductions (such as PF and tax). The system then processes this data to calculate the net salary using predefined formulas, ensuring accuracy and consistency.

The application is developed using Core Java concepts such as classes, objects, methods, loops, and conditional statements. It can be implemented as a console-based application and can be further enhanced into a graphical user interface using Java Swing or integrated with a database for large-scale use.

The system not only reduces manual workload but also minimizes calculation errors and improves the efficiency of payroll processing. It generates a well-structured payslip that includes all salary details, making it easy for both employers and employees to understand the breakdown of earnings and deductions.

Overall, this project demonstrates the practical application of Java programming in solving real-world business problems and highlights the importance of automation in improving organizational efficiency and accuracy.

1. INTRODUCTION

In today's modern business environment, efficient management of employee salaries is essential for the smooth functioning of any organization. Payroll processing is one of the most important administrative tasks, as it involves calculating employee earnings, managing allowances and deductions, and ensuring timely payment of salaries. A well-organized payroll system not only helps in maintaining employee satisfaction but also ensures compliance with financial and organizational policies.

Traditionally, many small and medium-sized organizations rely on manual methods or basic spreadsheet tools to handle payroll operations. These methods often require significant time and effort and are highly prone to human errors such as incorrect calculations, missing data, and inconsistencies in records. As the number of employees increases, managing payroll manually becomes increasingly complex and inefficient.

With the advancement of computer technology, automated payroll systems have become a necessity rather than a luxury. A computerized payroll system simplifies salary calculations, reduces manual intervention, and ensures accuracy in processing. It also helps in maintaining proper records, which can be accessed easily whenever required. Automation not only improves efficiency but also enhances transparency and reliability in payroll management.

The **Salary Payroll Processing and Payslip Generation System** is developed as a Java-based application to address these challenges. The system provides a structured and systematic approach to managing employee salary details. It allows the user to input employee information, define salary components such as basic pay, allowances (HRA, DA), and deductions (PF, tax), and automatically calculates the net salary.

One of the key features of this system is the generation of a detailed payslip, which provides a clear breakdown of the employee's earnings and deductions. This helps both the employer and employee understand how the final salary is calculated. The system is designed to be user-friendly, making it easy for users with basic computer knowledge to operate it efficiently.

The application is implemented using Core Java concepts such as classes, objects, methods, loops, and conditional statements. It can function as a console-based application and can be further enhanced into a graphical user interface using Java Swing or integrated with a database for storing employee records in larger organizations.

The main goal of this project is to reduce manual workload, eliminate calculation errors, and improve the overall efficiency of payroll processing. It also serves as a practical example of how programming concepts can be applied to solve real-world business problems.

1.1 Objective

To develop a Java-based payroll system

- To automate salary calculation
- To generate employee payslips
- To reduce manual errors
- To maintain employee salary records

1.2 Overview

The **Salary Payroll Processing and Payslip Generation System** is a Java-based application designed to provide a simple, efficient, and automated solution for managing employee payroll activities. The system acts as a digital platform where users can perform all payroll-related operations in a structured and organized manner.

The application begins by allowing the user to enter essential employee details such as employee name, ID, and other relevant information. After capturing employee data, the system enables the user to input various salary components, including basic salary, allowances such as House Rent Allowance (HRA) and Dearness Allowance (DA), and deductions such as Provident Fund (PF) and tax.

Once all the necessary data is entered, the system automatically processes the information and calculates the net salary using predefined formulas. This eliminates the need for manual calculations and significantly reduces the chances of errors. The automated calculation ensures consistency and accuracy in payroll processing.

A key feature of the system is its ability to generate a detailed and well-formatted payslip. The payslip includes a clear breakdown of all salary components, such as basic pay, allowances,

deductions, and the final net salary. This helps in maintaining transparency and allows employees to easily understand how their salary is computed.

The system is designed with a user-friendly interface, making it easy to operate even for users with minimal technical knowledge. It can be implemented as a menu-driven console application, where users can navigate through different options such as entering employee details, calculating salary, and generating payslips.

Additionally, the system is flexible and can be enhanced further based on requirements. It can be upgraded to include features such as:

- Storing multiple employee records
- Generating monthly payroll reports
- Integrating with databases like MySQL
- Developing a graphical user interface using Java Swing

Overall, the system simplifies payroll management by automating repetitive tasks, improving accuracy, and saving time. It serves as a practical example of how Java programming can be used to develop real-world applications that support business operations efficiently.

1.3 Features

The **Salary Payroll Processing and Payslip Generation System** provides several important features that help automate and simplify payroll management. These features are designed to improve accuracy, reduce manual work, and ensure efficient handling of employee salary data.

1. Employee Details Management

The system allows the user to store and manage employee information such as employee name, ID, and basic details. This helps in maintaining organized records and makes it easy to retrieve employee data whenever required.

2. Salary Component Management

The application enables the user to enter different salary components, including:

- Basic Salary
- House Rent Allowance (HRA)
- Dearness Allowance (DA)

This ensures that all parts of the salary structure are properly recorded and managed.

3. Deduction Handling

The system supports the inclusion of various deductions such as:

- Provident Fund (PF)
- Income Tax
- Other deductions (if needed)

These deductions are automatically subtracted from the total earnings to calculate the net salary accurately.

4. Automatic Salary Calculation

One of the core features of the system is automatic calculation of net salary. The system computes the final salary using the formula:

Net Salary = (Basic + Allowances) − Deductions

This eliminates manual calculations and reduces errors.

5. Payslip Generation

The system generates a well-structured payslip that includes:

- Employee details
- Salary breakdown

- Deductions
- Net salary

This payslip provides a clear and professional summary of the employee's earnings.

6. User-Friendly Interface

The application is designed as a simple menu-driven system, making it easy to use even for beginners. Users can navigate through options like entering data, calculating salary, and generating payslips without difficulty.

7. Time Efficiency

By automating payroll processes, the system saves significant time compared to manual methods. Salary calculations and payslip generation can be completed quickly.

8. Accuracy and Reliability

The system reduces human errors in calculations and ensures that salary computations are consistent and reliable.

9. Record Maintenance

All employee and salary details can be maintained systematically, which helps in future reference and reporting.

10. Scalability and Enhancement

The system can be extended with additional features such as:

- Database integration (MySQL)
- Multiple employee management
- GUI using Java Swing
- Monthly salary reports

2.PROGRAM ANALYSIS

2.1 Problem Definition

Manual payroll systems face issues like:

- Incorrect salary calculations
- Time-consuming processing
- Difficulty in managing records

2.2 Project Overview

This system:

- Stores employee data
- Calculates salary using formulas
- Generates a formatted payslip

2.3 Scopes and Limitations

Scope of the System

The **Salary Payroll Processing and Payslip Generation System** is designed to handle essential payroll operations in an efficient and systematic manner. The scope of the system defines what the application can perform and where it can be effectively used.

The key scope of the system includes:

- **Employee Salary Management**
  The system allows users to manage employee details and salary information, including basic pay, allowances, and deductions.
- **Automated Salary Calculation**
  It performs automatic calculation of net salary based on predefined formulas, ensuring accuracy and consistency.

- **Payslip Generation**

  The system generates a structured and easy-to-understand payslip showing salary breakdown, deductions, and final net salary.

- **Reduction of Manual Work**

  By automating payroll tasks, the system significantly reduces manual effort and saves time.

- **Improved Accuracy**

  The system minimizes human errors in calculations and ensures reliable payroll processing.

- **Record Maintenance**

  It helps in maintaining organized records of employee salary details for future reference.

- **Suitable for Small and Medium Organizations**

  The system is ideal for small businesses, startups, and educational projects where basic payroll processing is required.

- **Educational Purpose**

  The project demonstrates the practical application of Core Java concepts such as object-oriented programming, loops, and conditional statements.

- **Future Enhancements Possibility**

  The system can be expanded to include advanced features such as database integration, multi-user access, and graphical user interface.

2.4 **Limitations of the System**

While the system provides useful functionality, it also has certain limitations, especially in its basic version:

- **Limited to Console-Based Interface**

  The system may not have an advanced graphical interface, making it less visually interactive.

- **No Database Integration**

  Data is not permanently stored; employee details are lost once the program is closed.

- **Single User Operation**

  The system is designed for use by one user at a time and does not support multi-user access.

- **Basic Salary Structure Only**

  It handles only simple salary components and may not support complex payroll rules used in large organizations.

- **No Real-Time Updates**

  The system does not support real-time updates or synchronization with other systems.

- **Limited Security Features**

  There is no authentication or data security mechanism such as login systems.

- **Not Suitable for Large Enterprises**

  Large organizations with complex payroll requirements may need more advanced systems with database and cloud integration.

- **Manual Data Entry Required**

  All inputs must be entered manually, which can still take time for large numbers of employees.

3. SYSTEM DESIGN

Modules:

1. **Employee Module**
    - Stores employee name, ID
2. **Salary Module**
    - Basic Salary
    - Allowances (HRA, DA)
    - Deductions (PF, Tax)
3. **Calculation Module**
    - Calculates total salary
4. **Payslip Module**
    - Displays formatted payslip

4.METHODOLOGY

The methodology describes the systematic approach followed to design, develop, and implement the **Salary Payroll Processing and Payslip Generation System**. It ensures that the project is developed in a structured manner, starting from understanding the problem to delivering the final solution.

The development of this system follows a step-by-step process to ensure accuracy, efficiency, and proper functionality.

4.1 Problem Analysis

In this phase, the existing payroll system is studied to identify its drawbacks and challenges. Many organizations use manual or semi-manual payroll systems, which lead to:

- Errors in salary calculation
- Time-consuming processes
- Difficulty in managing employee records
- Lack of accuracy and consistency

Based on these issues, the requirements for an automated payroll system are identified. The system should be capable of handling employee data, calculating salaries, and generating payslips efficiently.

4.2 System Design

This stage involves planning the structure and workflow of the system before actual coding begins.

The system is divided into logical modules such as:

- Employee Management Module
- Salary Calculation Module
- Deduction Management Module
- Payslip Generation Module

The input, processing, and output flow are clearly defined to ensure smooth interaction between different components of the system.

4.3 Program Development

In this phase, the actual coding of the system is carried out using the Java programming language.

The development includes:

- Creating classes and methods
- Implementing input handling using Scanner
- Writing logic for salary calculation
- Designing the payslip output format

Core Java concepts used:

- Classes and Objects
- Conditional Statements (if-else)
- Loops
- Methods

This phase converts the system design into a working application.

4.4 Testing and Validation

Testing is performed to ensure that the system works correctly and produces accurate results.

Different test cases are used, such as:

Test Case	Expected Result
Enter employee details	Data stored correctly
Input salary components	Correct values accepted
Calculate salary	Accurate net salary
Invalid input	Error handled properly

This phase ensures that the system is free from errors and functions as expected.

4.5 Result and Documentation

After successful testing, the system produces the final output in the form of a payslip.

The results include:

- Employee details
- Salary breakdown
- Net salary

Proper documentation is prepared to describe:

- System design
- Implementation
- Working process

This helps in understanding and presenting the project clearly.

4.6 Algorithm

The algorithm provides step-by-step instructions for the system:

1. Start the program
2. Enter employee details
3. Input salary components (basic, allowances, deductions)
4. Calculate total earnings
5. Subtract deductions
6. Compute net salary
7. Generate payslip
8. Display output
9. End the program

4.7 Flowchart

The flowchart represents the working of the system in a visual form.

Flow:

Start

↓

Enter Employee Details

↓

Enter Salary Details

↓

Calculate Net Salary

↓

Generate Payslip

↓

Display Output

↓

End

You can draw this using:

- Draw.io
- Lucidchart
- MS Visio

5.SOURCE CODE

```
import java.util.Scanner;


public class PayrollSystem {


    String employeeName;
    int    employeeId;
    double basicSalary;
    int    daysWorked;


    double grossPay;
    double houseAllowance;
    double taxAmount;
    double providentFund;
    double netPay;


    void inputEmployeeData(Scanner sc) {
        System.out.println("===== Enter Employee Details =====");


        System.out.print("Employee Name   : ");
        employeeName = sc.nextLine();
```

```
    System.out.print("Employee ID     : ");
    employeeId = sc.nextInt();

    System.out.print("Basic Salary (monthly, for 30 days): Rs. ");
    basicSalary = sc.nextDouble();

    System.out.print("Days Worked this month: ");
    daysWorked = sc.nextInt();
    sc.nextLine();
}

void calculateSalary() {
    grossPay = (basicSalary / 30.0) * daysWorked;

    houseAllowance = 1500.0;

    taxAmount     = grossPay * 0.10;
    providentFund = grossPay * 0.05;

    netPay = grossPay + houseAllowance - taxAmount - providentFund;

    grossPay      = Math.round(grossPay      * 100.0) / 100.0;
    taxAmount     = Math.round(taxAmount     * 100.0) / 100.0;
    providentFund = Math.round(providentFund * 100.0) / 100.0;
    netPay        = Math.round(netPay        * 100.0) / 100.0;
```

```
}

void generatePayslip() {
    System.out.println();
     System.out.println("========================================");
     System.out.println("          EMPLOYEE PAYSLIP            ");
     System.out.println("========================================");
     System.out.printf("  Name         : %s%n", employeeName);
     System.out.printf("  Employee ID  : %d%n", employeeId);
     System.out.printf("  Days Worked  : %d / 30%n", daysWorked);
     System.out.println("----------------------------------------");
     System.out.println("  EARNINGS");
     System.out.printf("    Gross Pay        : Rs. %8.2f%n", grossPay);
     System.out.printf("    House Allowance  : Rs. %8.2f%n", houseAllowance);
     System.out.println("----------------------------------------");
     System.out.println("  DEDUCTIONS");
     System.out.printf("    Income Tax (10%%) : Rs. %8.2f%n", taxAmount);
     System.out.printf("    Provident Fund(5%%): Rs. %8.2f%n", providentFund);
     System.out.println("----------------------------------------");
     System.out.printf("  NET PAY             : Rs. %8.2f%n", netPay);
     System.out.println("========================================");
     System.out.println("     Thank you! Have a great month.     ");
     System.out.println("========================================");
 }
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        PayrollSystem payroll = new PayrollSystem();

        payroll.inputEmployeeData(sc);
        payroll.calculateSalary();
        payroll.generatePayslip();

        sc.close();
    }
}
```

6. INPUT AND OUTPUT SCREENS

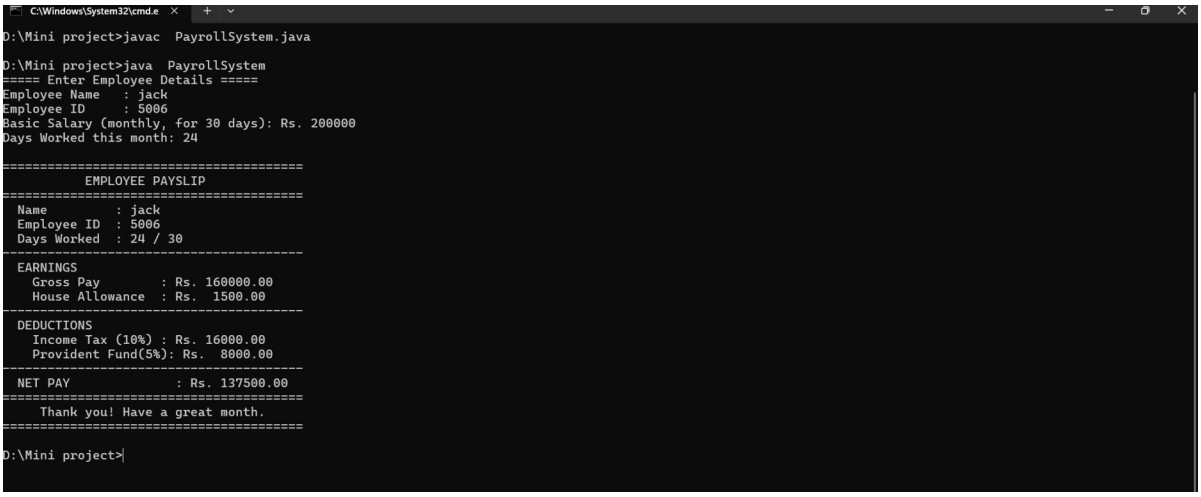
INPUT SCREEN:



OUTPUT SCREEN:



FINAL OUTPUT:



7. DISCUSSION & CONCLUSION

7.1 Discussion

The **Salary Payroll Processing and Payslip Generation System** was developed to automate the process of calculating employee salaries and generating payslips in an efficient and accurate manner. The system successfully addresses the problems associated with manual payroll processing, such as calculation errors, time consumption, and difficulty in maintaining records.

During the development of this project, various Core Java concepts such as classes, objects, methods, loops, and conditional statements were applied. The program was structured into separate modules for input handling, salary calculation, and payslip generation, which improved the readability and maintainability of the code.

The system calculates the salary based on the number of days worked, which makes it more practical and realistic for real-world scenarios. It also includes additional components such as house allowance, income tax, and provident fund deductions. These features ensure that the payroll system reflects actual business practices.

The use of formatted output (using printf) enhances the presentation of the payslip, making it clear and professional. The generated payslip provides a detailed breakdown of earnings and deductions, which improves transparency and helps users understand the salary structure.

The system was tested with different inputs, including varying salaries and working days, to verify its accuracy and reliability. The results showed that the system performs calculations correctly and consistently. It also handles user input effectively, ensuring smooth operation.

However, the system has certain limitations. It currently supports only a single employee at a time and does not store data permanently. Additionally, it is implemented as a console-based application, which may not be as user-friendly as graphical systems.

Despite these limitations, the project successfully demonstrates how Java programming can be used to develop a practical payroll management system. It provides a strong foundation that can be further enhanced with advanced features such as database integration, graphical user interface, and multi-user support.

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7.2 Conclusion

The **Salary Payroll Processing and Payslip Generation System** has been successfully designed and implemented using Java programming. The project achieves its primary objective of automating salary calculation and generating accurate payslips, thereby reducing manual effort and minimizing errors.

The system simplifies payroll processing by allowing users to input employee details, calculate salary based on working days, and generate a detailed payslip that includes earnings, deductions, and net pay. This improves efficiency, accuracy, and transparency in payroll management.

Through this project, the practical application of Core Java concepts such as object-oriented programming, methods, loops, and conditional statements has been clearly demonstrated. It also highlights the importance of automation in modern business environments.

Although the current system is limited to a console-based application and lacks features such as data storage and multi-user support, it provides a solid base for further development. Future enhancements can include:

- Integration with databases for storing employee records
- Development of a graphical user interface using Java Swing
- Implementation of advanced payroll features such as bonuses and tax slabs
- Support for multiple employees and report generation

In conclusion, this project not only fulfills its intended purpose but also serves as a valuable learning experience in software development. It demonstrates how programming skills can be effectively used to solve real-world problems and improve organizational efficiency.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java

• https://www.geeksforgeeks.org

• https://www.w3schools.com

• https://www.javatpoint.com